

Somaiya Vidyavihar University K.J.
Somaiya School of Engineering

Batch: A2 Roll No.: 16010324044

Experiment / assignment / tutorial No. 01

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

Title: Title: Amplitude Shift Keying

Objectives:

1. To implement amplitude shift keying (ASK) digital modulation technique (transmitter)
2. To write a program for ASK in MATLAB

OUTCOME: Analyse different band pass digital transmission and reception of signal

Draw Circuit Diagram:

Write specifications of IC LF398: —?

Draw pin diagram of IC LF398:

Procedure:

- Assemble the circuit as shown in circuit diagram
- Apply digital modulating signal and analog carrier signal with given frequency and amplitude.
- Observe the ASK output on DSO and tabulate the readings.
- Draw the waveforms of modulating, carrier and ASK signal.
- Write a program in MATLAB to observe the ASK waveforms.

Observation Table:

Modulating Signal		Carrier signal		BASK Signal	
Amplitude	Frequency	Amplitude	Frequency	Amplitude for logic 0 Input	Amplitude for logic 1 Input
+5V	1KHz	+2.5V	10KHz	0	+2.5V

Matlab Code:

```
clc; %clear command window
clear; %clear workspace
close all; %close all open figure windows
%% Input Binary Data
data = [1 0 1 1 0 1 0 0];
%% Parameters
A = 5; % Carrier Amplitude
fc = 10; % Carrier Frequency (Hz)
Tb = 1; % Bit Duration (s)
fs = 1000; % Sampling Frequency (Hz)
```

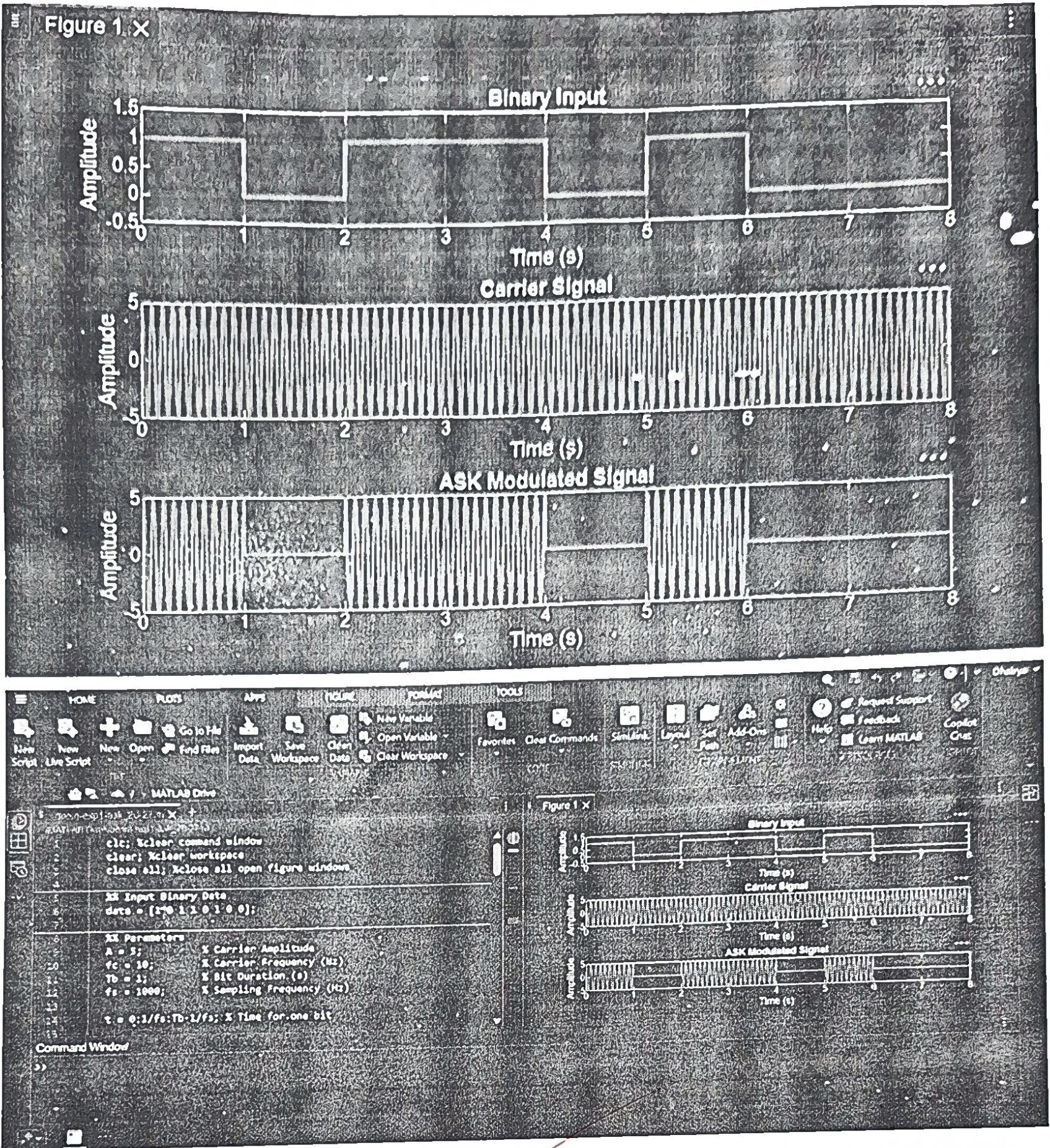


```

t = 0:1/fs:Tb-1/fs; % Time for one bit
%% Generate NRZ Signal
nrz = [];
for i = 1:length(data)
    nrz = [nrz data(i)*ones(1,length(t))];
end
%% Total Time
total_time = 0:1/fs:length(data)*Tb-1''s;
%% Carrier Signal
carrier = A*cos(2*pi*fc*total_time);
%% ASK Signal
ask = [];
for i = 1:length(data)
    if data(i)==1
        ask = [ask A*cos(2*pi*fc*t)];
    else
        ask = [ask zeros(1,length(t))];
    end
end
%% Plotting
figure;
subplot(3,1,1);
plot(total_time,nrz,'LineWidth',2);
axis([0 length(data) -0.5 1.5]);
grid on;
title('Binary Input');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,2);
plot(total_time,carrier,'LineWidth',1.5);
grid on;
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(3,1,3);
plot(total_time,ask,'LineWidth',1.5);
grid on;
title('ASK Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');

```

Matlab Output:



Conclusion:

In this experiment, I successfully implemented ASK modulation in MATLAB. The generated waveform shows that the carrier signal is transmitted only when the input bit is 1, while no carrier is transmitted for input bit 0. Thus, the MATLAB simulation verifies the working principle of Binary Amplitude Shift Keying (BASK).

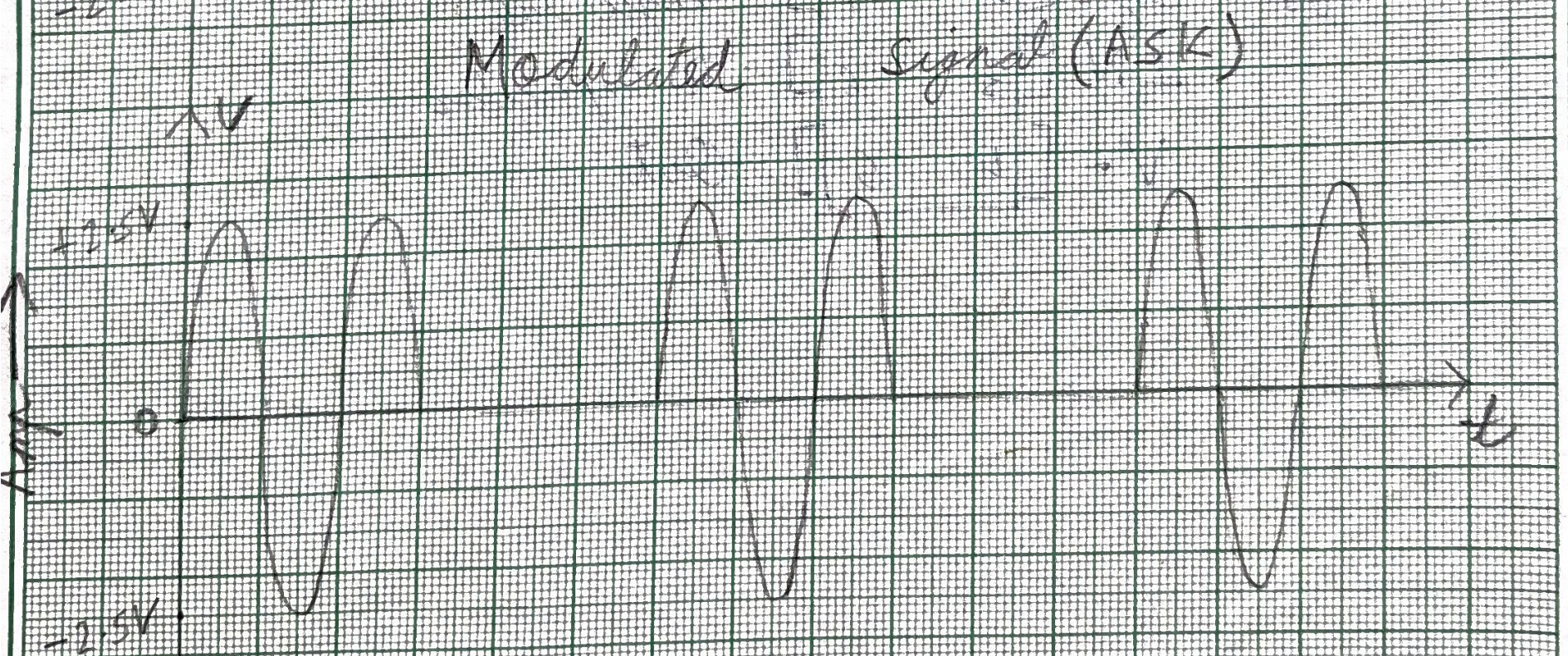
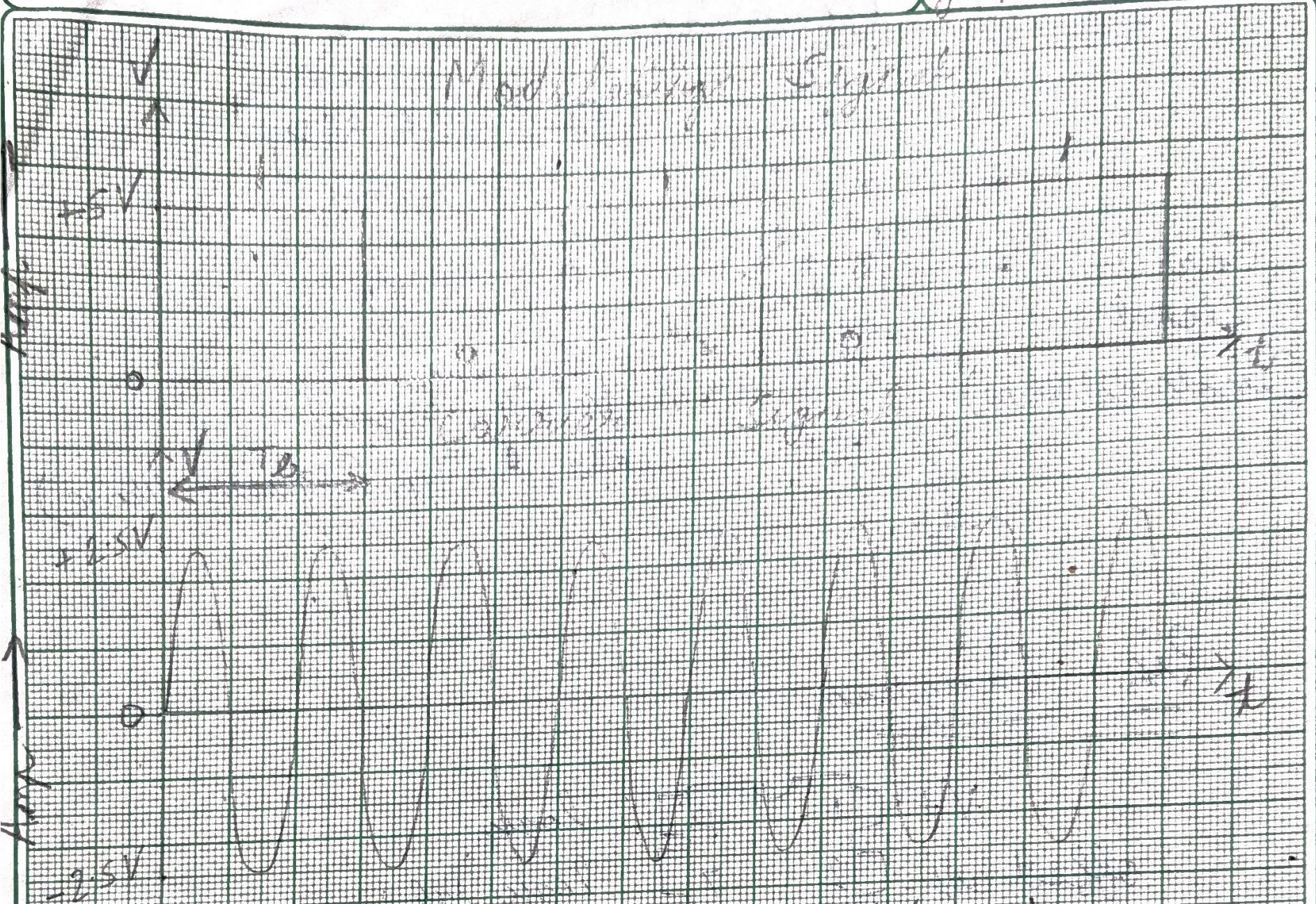
Signature 4/18/26
Signature of faculty in-charge

EXP 1

SCALE :-

x : 3 Units = 1 μ s

y : 1 Unit = 1 Volt



21/07/25